Predicting Emergency Room Wait Times Using Machine Learning

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Abstract

Prolonged emergency room (ER) wait times are strongly associated with patient dissatisfaction, delayed treatment, and hospital inefficiency, making wait-time prediction an important operational problem. This project investigates whether supervised machine learning can accurately predict ER wait times and, more importantly, which feature types provide reliable predictive signal. Following the original proposal, the work was organized into two phases. Phase 1 used the *Hospital Emergency Room Dataset* (9,216 encounters), a patient-level dataset containing demographic and administrative variables such as age, gender, race, referral department, admission status, and visit timestamp information. A broad suite of regression models was evaluated, including Linear Regression, Ridge, Lasso, ElasticNet, Decision Tree, Random Forest, Extra Trees, Gradient Boosting, XGBoost, LightGBM, Voting Regressor, and Stacking Regressor. Despite this extensive model comparison, predictive performance remained extremely weak, with the best validation score only $R^2 = 0.008$ and negative test-set performance, indicating that patient demographics and simple visit descriptors contain little useful information for predicting wait time.

Guided by this result, Phase 2 shifted to the *ER Wait Time Dataset* (5,000 records), which contains richer operational variables such as urgency level, nurse-to-patient ratio, specialist availability, facility size, time of day, and day of week. Initial experiments produced nearly perfect accuracy, but further investigation revealed severe target leakage: several process-time variables directly encoded components of the total wait time. After removing the leaked features, model performance became realistic and interpretable. The best trustworthy model, Gradient Boosting, achieved $R^2 = 0.754$ on validation and $R^2 = 0.846$ on the held-out test set, with $\text{MAE} = 19.27$ minutes and $\text{RMSE} = 26.73$ minutes. Feature importance analysis showed that urgency level, time of day, day of week, and nurse-to-patient ratio were the dominant predictors. Overall, the project demonstrates that operational context is substantially more informative than demographics for ER wait-time prediction, while also highlighting the importance of rigorous leakage detection in healthcare machine learning.

1. Introduction

Emergency Departments (EDs) are among the most operationally constrained units in healthcare. Long waiting times increase patient stress, delay treatment, reduce satisfaction, and contribute to *left without being seen* (LWBS) events, where patients leave before receiving medical care. Because wait time is influenced by dynamic factors such as patient demand, acuity distribution, staffing, and service bottlenecks, hospitals need predictive tools that go beyond historical averages.

The central goal of this project was to evaluate whether machine learning regression can predict ER wait times accurately and to determine which types of features are most useful. The proposal hypothesized that operational and temporal variables would outperform purely demographic information. To test this idea, the project followed a staged plan: begin with a patient-level baseline dataset, then move to a richer operational dataset if the first phase showed limited predictive signal.

2. Method

2.1 Datasets

Three datasets were proposed, and two were implemented in the final project.

Dataset 1: Hospital Emergency Room Dataset:
This dataset contains 9,216 patient encounters with variables such as age, gender, race, referral department, admission status, case-management indicator, patient satisfaction score, admission date, and wait time. It is primarily demographic and administrative in structure and therefore served as the baseline dataset.

Dataset 3: ER Wait Time Dataset:
This dataset contains 5,000 records and includes richer operational variables such as urgency level, day of week, season, time of day, nurse-to-patient ratio, specialist availability, and facility size. These features better reflect the real factors that shape congestion and patient flow in an ER.

Dataset 2: Simulated ER Wait Time Dataset:
Although included in the proposal, this dataset was not incorporated into the final experiments. As a result, the originally proposed multi-dataset stacking phase was only partially completed.

2.2 Preprocessing

For Dataset 1, direct identifiers and low-value fields were removed, missing categorical values were imputed, and temporal features were extracted from the admission date. Categorical variables were one-hot encoded and age was standardized. The data was ordered chronologically, and the top 3% of training wait-time outliers were removed to stabilize fitting. The split strategy used a held-out test set plus a separate validation partition.

For Dataset 3, additional time-based features were extracted from visit date, and preprocessing was handled using a full `ColumnTransformer` pipeline with scaling for numeric variables and one-hot encoding for categorical variables. Variables representing post-visit outcomes or process durations were audited carefully because they risked leaking information about the target.

2.3 Models and Evaluation

The project compared a wide range of supervised regression models. In Phase 1, these included Linear Regression, Lasso, Ridge, ElasticNet, Decision Tree, Random Forest, Extra Trees, Gradient Boosting, XGBoost, LightGBM, Voting Regressor, and Stacking Regressor. In Phase 2, the evaluation focused on DummyRegressor, Linear Regression, Random Forest, and Gradient Boosting.

Performance was measured using R^2, MAE, and RMSE, which aligns with the proposal's requirement for interpretable regression metrics. Cross-validation and residual analysis were also used to evaluate generalization and identify error patterns.

3. Experiments

Phase 1: Baseline modeling on Dataset 1:
The first experimental phase tested whether patient-level features alone could support meaningful ER wait-time prediction. A broad model comparison was performed, together with time-aware validation using `TimeSeriesSplit`. Feature importance from the tree-based models suggested that age and simple temporal features had the greatest influence, but the overall predictive signal remained negligible.

Phase 2: Operational modeling on Dataset 3:
Because the baseline phase performed poorly, the project moved to the richer operational dataset. Initial results were almost perfect, which immediately suggested that the model might be benefiting from leaked information. Feature importance inspection confirmed that three process-time variables—`Time to Registration`, `Time to Triage`, and `Time to Medical Professional`—were effectively encoding the target `Total Wait Time`. These features were removed, and the models were retrained on the corrected no-leak feature set.

4. Results

Table 1 summarizes the most important results from both phases.

Phase	Model / setting	R^2	MAE	RMSE
1	Gradient Boosting (validation)	0.008	12.23	14.13
1	Random Forest (test)	-0.075	12.99	15.18
2	Dummy baseline	-0.070	51.78	59.82
2	Linear Regression (with leakage)	1.000	0.00	0.00
2	Random Forest (with leakage)	0.999	0.98	1.55
2	Gradient Boosting, no-leak (validation)	0.754	20.61	28.71
2	Random Forest, no-leak (validation)	0.751	20.59	28.86
2	Gradient Boosting, no-leak (test)	0.846	19.27	26.73

The results show three clear outcomes. First, Dataset 1 failed to provide meaningful predictive power. Even with extensive model comparison, the best validation performance was near zero, which indicates that demographics and basic visit descriptors are not enough to explain ER waiting time. Second, Dataset 3 contained substantial predictive signal, but the initial perfect accuracy was invalid because of target leakage. Third, after leakage removal, the best model remained strong and clinically plausible, demonstrating that operational variables are far more informative than patient-level features.

Feature-importance analysis for the no-leak Gradient Boosting model showed that the strongest predictors were urgency level, time of day, day of week, and nurse-to-patient ratio. These variables align with real hospital workflow and support the project's original hypothesis.

5. Conclusion

This project evaluated whether machine learning can predict ER wait times and which feature types make such prediction reliable. The findings strongly support the conclusion that patient demographics alone are weak predictors, while operational context is highly informative. The first dataset produced near-zero predictive performance despite the use of many regression models, confirming that basic patient-level variables do not capture the mechanisms driving ER congestion. In contrast, the second implemented dataset achieved strong performance after leakage was removed, with the best trustworthy model—Gradient Boosting—reaching $R^2 = 0.846$ on the test set.

Beyond predictive performance, the most important lesson from the project is methodological. In healthcare machine learning, apparently perfect results can be misleading if the feature set contains leaked information from downstream processes. The project therefore succeeded not only by producing a strong model, but also by demonstrating the importance of careful feature auditing, realistic evaluation, and critical interpretation of results.

Several limitations remain. Dataset 2 was not integrated, and the full stacked multi-dataset framework proposed at the start was not completed. In addition, residual analysis suggests that extreme long-wait cases remain difficult to predict. Future work should incorporate the missing dataset, test quantile or transformed-target regression to better model long waits, and apply stricter time-aware validation for deployment-oriented evaluation.

References

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